

KING COUNTY

House Price Prediction

A regression pipeline for residential sale prices — comparing Linear Regression, KNN and Random Forest

Ironhack Data Science & Machine Learning Bootcamp — Team Project

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Project Overview

Predicting residential sale prices in King County, WA (Seattle area) from property characteristics.

21,613

property sales
(May 2014 – May 2015)

21

original features
per property

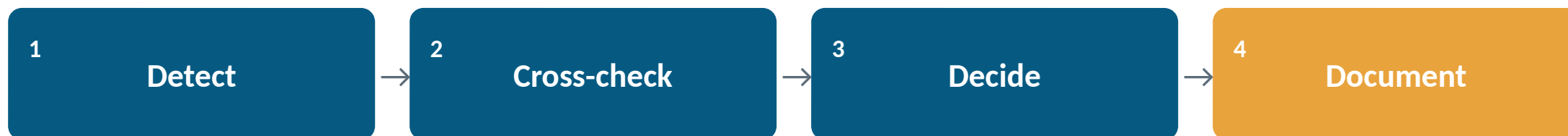
3

models compared
(Linear, KNN, Random Forest)

Assigned model: Random Forest Regressor

Linear Regression and KNN were built as baselines to test whether the price–feature relationship is linear before committing to a more complex model.

Data Cleaning: A Repeatable Process



Never removed data based on a single number in isolation — every anomaly was checked against a related variable first.

33 bedrooms, 1,620 sqft

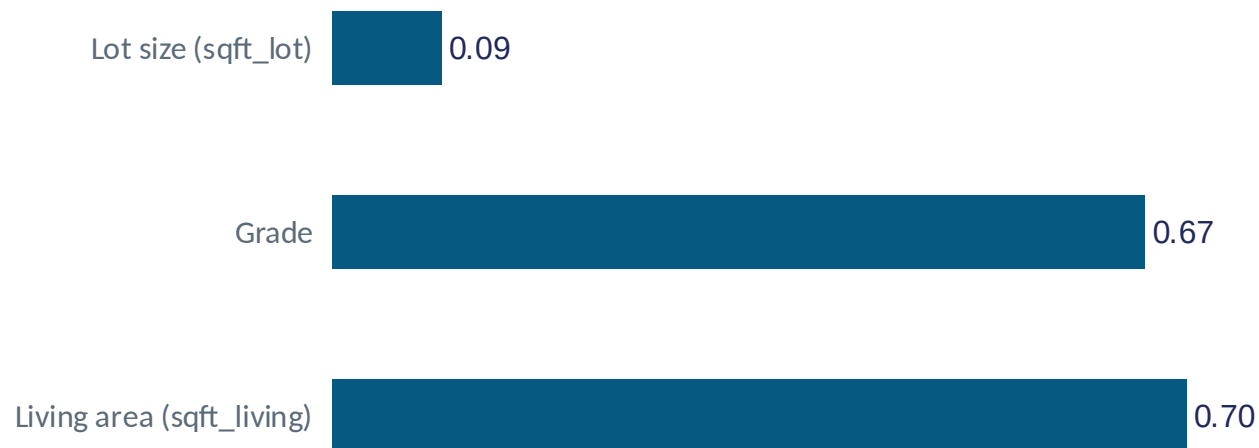
Physically impossible for that size — confirmed via sqft_living, not assumed. Row dropped.

“Grade” scale mismatch

Documentation said 1–11, but data showed a consistent 3–13 range (102 rows), backed by price & sqft trends. Data trusted over outdated docs — full range kept.

What Actually Drives Price?

Pearson correlation with price, from exploratory data analysis

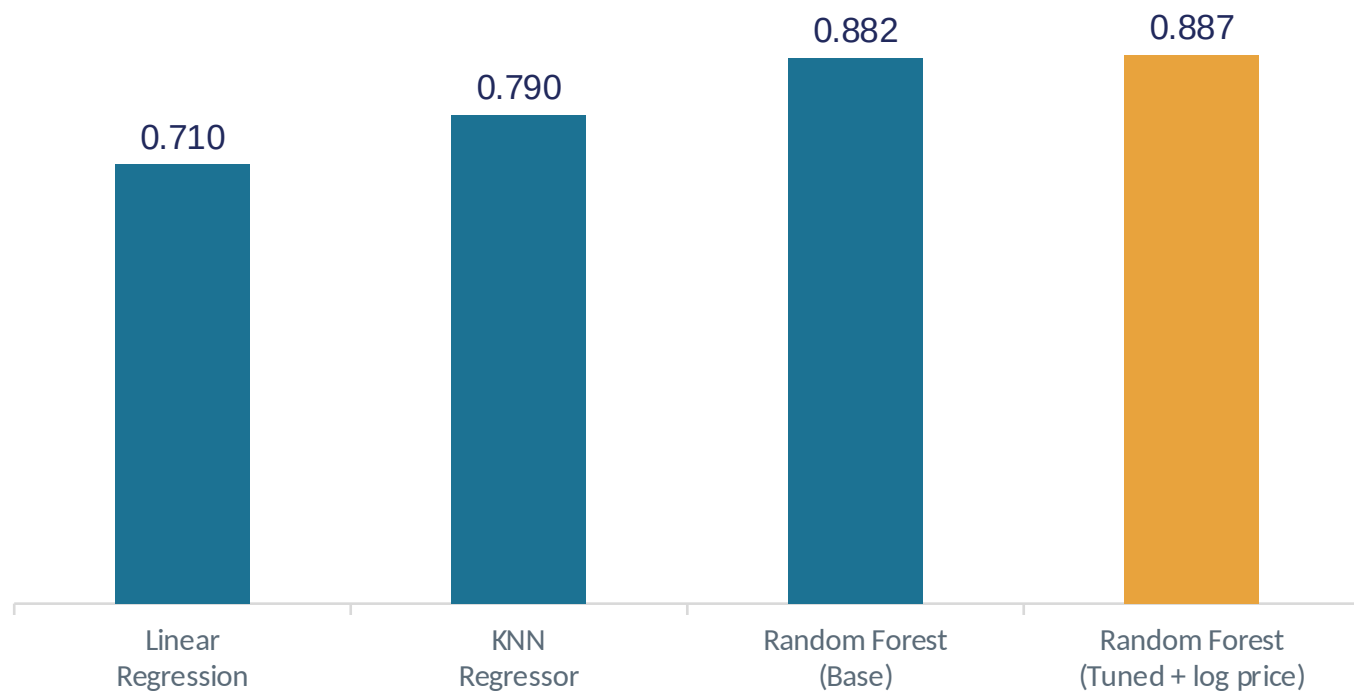


Key insight

sqft_living and grade correlate strongly with price. Lot size barely matters on its own — size of the home, not the plot, is what buyers pay for.

Feature engineering: sale_year / sale_month extracted from date · was_renovated + years_since_renovation replace a misleading yr_renovated · zipcode label-encoded (kept lat/long for continuous geography)

From Linear Baseline to Tuned Random Forest



Final model

R² 0.887

MAE \$68,493.53

Log-transforming price (best hyperparameters: max_depth=30, n_estimators=300) reduced the train/test overfitting gap from 0.098 to 0.085 — better generalization, at a small cost in dollar-term MAE.

KNN beating Linear Regression was the first signal the price-feature relationship isn't linear — confirmed decisively by Random Forest.

What the Model Actually Learned

Top drivers of predicted price, from the tuned Random Forest's feature importances

-  **Living area (sqft_living)**
Largest single driver — consistent with its strong linear correlation.
-  **Grade**
Second largest — also consistent with the correlation analysis.
-  **Latitude**
Ranks far higher than its weak linear correlation would suggest: specific areas command premium prices independent of a simple geographic gradient — a non-linear pattern linear correlation can't detect, but Random Forest can.

Notably, zipcode_encoded ranked low despite representing location — Label Encoding gives it arbitrary numeric order, unlike the continuous signal in lat/long.

Key Takeaways



Data cleaning as a discipline, not a checklist — every anomaly cross-checked against related variables before deciding, never dropped on a single number.



Trusted the data's internal consistency over outdated documentation, verified empirically (the grade range).



Tested assumptions rather than accepting them — removing seemingly-redundant sqft columns was tried and measured, not just theorized.



Collaboration paid off directly: a teammate's log-transform finding generalized to this model, improving R^2 and reducing overfitting.

R^2 0.887

Final model: Random Forest (Tuned + log-transformed price)

github.com/madatascienceml/IronKaggle_Project